End-to-End Sales Forecasting & Demand Intelligence System

Executive Business Report

Executive Summary

This project was developed to support business managers, supply chain teams, and financial decision makers by predicting future product demand using historical sales information. The system combines time series forecasting, anomaly detection, demand segmentation, and interactive business dashboards into a single decision-support solution. Instead of relying only on historical averages, the system identifies trends, seasonal behavior, and unusual sales patterns to help businesses make better inventory and purchasing decisions.

Data Exploration and Business Understanding

The Superstore sales dataset was cleaned and prepared by converting date fields into proper datetime format, extracting useful time features such as year, month, quarter, and week, and checking for missing values and duplicate records. Daily sales were aggregated into weekly and monthly totals to support both exploratory analysis and forecasting. Visualizations highlighted long-term sales growth, seasonal demand, category performance, and regional differences.

Time Series Analysis

Monthly sales trends were decomposed into trend, seasonal, and residual components. The analysis revealed recurring seasonal demand together with long-term growth patterns. The Augmented Dickey-Fuller test was used to evaluate stationarity, and differencing was applied where necessary to improve forecasting performance. These preprocessing steps ensured that statistical forecasting models were trained on suitable time series data.

Forecasting Models

Three forecasting approaches were implemented and compared. SARIMA captured statistical relationships within the time series, Prophet modeled yearly and monthly seasonality, and XGBoost transformed historical observations into supervised learning features using lag variables and rolling averages. Each model was evaluated using MAE, RMSE, and MAPE to identify the most reliable forecasting approach.

Anomaly Detection

Weekly sales data was examined using Isolation Forest and Z-Score techniques to detect unusual sales spikes and unexpected drops. These anomalies may represent promotional campaigns, seasonal shopping events, supply chain disruptions, or unexpected market behavior. Identifying such events allows businesses to investigate the underlying causes and respond proactively.

Product Demand Segmentation

Product sub-categories were grouped into demand segments using K-Means clustering after feature scaling and dimensionality reduction through PCA. The resulting clusters represent products with similar sales characteristics. High-demand products require consistent inventory availability, seasonal products should be stocked before peak demand periods, and lower-demand products should be managed carefully to reduce storage costs.

Business Recommendations

1. Increase inventory planning before forecasted high-demand periods to reduce stock shortages.
2. Monitor anomalies continuously to identify unexpected business events and investigate operational issues.
3. Allocate inventory based on regional demand forecasts instead of applying identical stocking policies across all locations.
4. Retrain forecasting models regularly using newly available sales data to improve prediction accuracy.
5. Integrate dashboard insights into weekly management meetings to support data-driven business decisions.

Project Limitation

The forecasting system is based on historical sales information and therefore assumes future market conditions are reasonably similar to past behavior. Sudden economic changes, supply chain disruptions, or new competitor strategies may reduce prediction accuracy. Forecasts should therefore be used alongside business expertise rather than as the sole basis for strategic decisions.

Conclusion

This project demonstrates how modern data science techniques can improve retail decision making by combining forecasting, anomaly detection, clustering, and interactive visualization into a unified platform. The developed solution provides actionable insights that help organizations optimize inventory levels, improve customer satisfaction, reduce operational costs, and support long-term business growth.